

Aagosh Raj Srivastava

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TECHNICAL SKILLS

Languages: Java, JavaScript, TypeScript, Python, SQL, Kotlin, HTML/CSS

Backend & Frameworks: Node.js, Express, Spring Boot, Spring Cloud, Socket.IO, Fastify, REST APIs, WebSockets

Frontend: React 18+, Vite, Zustand, Three.js, Tailwind CSS, GSAP

Databases & Messaging: PostgreSQL, MongoDB, Redis, Apache Kafka, Prisma ORM

DevOps & Cloud: Docker, Kubernetes, GitHub Actions, Render, Vercel, Prometheus, Grafana

Security: Argon2id Hashing, JWT, OAuth, Zero-Trust Architecture, CORS, Rate Limiting

Relevant Coursework: Distributed Systems, Advanced DBMS, Cryptography & Network Security, Operating Systems, Computer Networks, Compiler Design, Deep Learning, Design & Analysis of Algorithms

FEATURED PROJECTS

ORBIT — Scalable Post-Quantum Messaging Platform

[GitHub](#)

- **Distributed Messaging Backbone:** Architected a high-concurrency real-time communication engine utilizing Socket.IO and a Redis adapter to enable horizontal scaling for 100+ concurrent group broadcasts with sub-100ms latency; engineered an offline-first queuing system with Redis persistence, idempotency deduplication, and automated replay to guarantee zero message loss.
- **Post-Quantum Zero-Trust Security:** Engineered a cutting-edge cryptographic defense layer integrating X3DH key agreement, Double Ratchet E2EE, Argon2id hashing with a 64-byte pepper, and ML-KEM-768 post-quantum resistance to successfully pass OWASP Cryptographic Failure threat modeling; built device-bound session management with hardware attestation and hierarchical protocol versioning (*v1 RSA* → *v4 Nexus Sender Key*).
- **Defensive Infrastructure & Controls:** Built a distributed defense-in-depth middleware stack (Helmet CSP, CORS validation, HPP) executing per-user/IP rate-limiting, 50-event socket thresholds, and a 90-point behavioral anomaly auto-blocking engine to mitigate bot attacks by 95%+; deployed a cryptographic gesture-based authentication system alongside graceful microservice shutdown mechanics to ensure clean container teardowns.

PRAHAR — Enterprise Geospatial Intelligence Platform

[GitHub](#)

- **Real-Time Geospatial Telemetry Engine:** Engineered an event-driven telemetry pipeline using Socket.io and a localized high-performance cache layer ('memoryEngine') to broadcast 2-second heatmap ticks across concurrent operational zones, slash alert detection latency from 15s to μ s, and maintain sub-millisecond lookup times. Integrated GPU-accelerated Three.js and WebGL rendering pipelines with frustum culling to visualize 100K+ live geospatial data points at a stable 60 FPS while minimizing memory overhead by 40%.
- **Predictive Forecasting & State Architecture:** Designed a hierarchical event state machine supporting multi-phase transitions (*active* → *terminated* → *restarted*) with automated asynchronous historical aggregation over seeded datasets. Built a predictive capacity forecasting engine computing guard dispatch eligibility within 30-second intervals via time-series analysis and Haversine distance calculations, immediately triggering automated, targeted emissions to mobile clients upon 95% capacity threshold breach anomalies.
- **High-Availability Infrastructure & Query Optimization:** Developed an $O(n)$ proximity-search query layer handling concurrent guard position updates with distance sorting and GEORADIUS-equivalent precision. Optimized database access patterns utilizing compound indexed lookups and 'lean()' projections on dense aggregations to guarantee sub-100ms response times; enforced enterprise-grade security via robust JWT role-validation middleware, standardized error boundaries, and centralized event emission patterns.

FORGE AI — Intelligent AI Code Generation Platform

[GitHub](#)

- **Autonomous AI Orchestration & Full-Stack SaaS Infrastructure:** Architected a dual-model pipeline routing between Qwen 32B/7B to optimize inference, cutting latency to sub-60s across 1,000+ daily requests with a 95% success rate and credit-safe refund logic. Paired a '@neondatabase/serverless' and Prisma backend to support 5K concurrent profiles with a multi-viewport UI, integrating morphdom-inspired DOM diffing for flicker-free streaming and one-click version rollbacks.

PROFESSIONAL EXPERIENCE

Finlatics — Generative AI Project Trainee

Jun 2025 — Aug 2025

- Optimized LLM response streaming UI components by tuning event-loop processing and implementing intelligent token buffering, reducing perceived user latency by 250ms and improving engagement metrics by 35%.
- Designed high-precision interfaces for NLP pipelines implementing prompt-chaining orchestration with real-time state visualization, increasing model accuracy feedback efficiency by 40%.

Future Interns — Full Stack Developer Intern

Jun 2025 — Jul 2025

- Accelerated Core Web Vitals (LCP) by 40% through aggressive code-splitting, lazy component loading, and optimized asset delivery, achieving 100/100 Lighthouse score on production application.
- Engineered high-conversion e-commerce interface using React and Stripe with idempotent payment processing for 99.9% transaction consistency. Reduced payment failure rate from 2.3% to 0.1%.

EDUCATION

SRM University, Andhra Pradesh — B.Tech Computer Science and Engineering (AI/ML) Aug 2023 – 2027